

4. What is the purpose of **System Calls** in relation to **User mode** and **Kernel mode** ? (0.75 pt)

- System calls allow user-mode programs to request services from the kernel without directly accessing hardware.
- System calls allow user programs to request kernel services safely.
- System calls form the safe interface between applications and the operating system.

5. What is a **Desktop Hypervisor** ? Explain its purpose and provide **examples of commonly used desktop hypervisors**. (1.5 pt)

A **desktop hypervisor** is a type of virtualization software that enables a personal computer to run **multiple operating systems concurrently** on the same hardware.

Desktop hypervisors are designed for **end-user machines**, focusing on **ease of use, testing, development, and experimentation**.

Desktop hypervisors are commonly used by developers, testers, and IT enthusiasts who need to run Linux, Windows, or other OSs simultaneously without rebooting.

Examples of desktop hypervisors include:

- **VirtualBox** (Oracle)
- **VMware Workstation** (VMware)
- **Parallels Desktop** (for macOS)

☐ **EXERCISE 2** (3 = 4*0.75 POINTS) : Answer the following questions by means of short answers.

Questions		Answers
01	03 examples of UNIX-Based systems	macOS; AIX (IBM); HP-UX; Tru64 UNIX ; Solaris (Oracle)
02	03 examples of UNIX-Like systems	MINIX Linux distributions: Ubuntu, Debian, Fedora, CentOS, Arch Linux BSD systems: FreeBSD, OpenBSD, NetBSD, DragonFly BSD
03	03 examples of Community-driven distributions	Debian – Entirely community-maintained; stable and widely used as a base for other distros. Arch Linux – Rolling-release distro focused on simplicity and user control. Gentoo – Source-based, highly customizable, maintained by community contributors. Slackware – One of the oldest Linux distributions, community-maintained.
04	03 examples of Company-backed distributions	Ubuntu (Canonical) – Popular desktop and server distro, backed by Canonical Ltd. Red Hat Enterprise Linux (RHEL) – Enterprise-focused, supported by Red Hat. SUSE Linux Enterprise (SLE) – Enterprise distro backed by SUSE. Fedora – Sponsored by Red Hat, community-supported but influenced by company direction.

☐ **EXERCISE 3 (2.5 = 10*0.25 POINTS) : True / False** – Indicate whether each statement is True (T) or False (F)

	Statements	True / False
01	BIOS and UEFI always allow the operating system to boot even if critical hardware components, such as RAM or storage devices, fail the initial diagnostic tests.	False
02	APIs allow programs to use system functions with a direct CPU interaction, and can provide access to hardware-related operations.	False
03	Some Linux distributions, whether community-driven or company-backed , do not follow the open-source philosophy. <i>All Linux distributions, whether community-driven or company-backed, follow the open-source philosophy.</i>	False
04	An upgrade installs small patches or fixes to an existing software version, while an update replaces the current software with a newer major version.	False
05	The Virtualization mimics hardware so that software for one platform can run on a completely different platform. <i>Virtualization allows multiple operating systems to run on the same hardware without simulating hardware, whereas emulation mimics hardware so that software for one platform can run on a completely different platform.</i>	False
06	The GUI remains consistent across different Linux distributions, while the CUI can vary significantly depending on the chosen desktop environment. <i>The console experience is consistent everywhere, but the graphical experience can radically change depending on the desktop environment.</i>	False
07	WSL allows Linux applications to access Windows files and vice versa easily.	True
08	A regular release of a Linux distribution receives updates and security patches for a longer period than a LTS version. <i>A Long-Term Support (LTS) version of a Linux distribution receives updates and security patches for a longer period than a regular release, which typically has a shorter support cycle.</i>	False
09	The file <code>~/.bashrc</code> is a personal configuration file for the Bash shell.	True
10	Environment variables are only used by the kernel and never by user applications. <i>User applications also use environment variables, not just the kernel.</i>	False

☐ **EXERCISE 4 (2.5 = 10*0.25 POINTS) : SCQ / MCQ** - Check all correct statements.

01	Which of the following statements best describes a difference between Monolithic and Microkernel architectures?					
	Monolithic kernels run most OS services in user space, while Microkernels run all services in kernel space.	☒	Microkernels run only essential services in kernel space, improving reliability compared to Monolithic kernels.	☒	Monolithic kernels are slower because they use inter-process communication for every service.	Both architectures execute all services in kernel mode for performance reasons.
02	The CPU switches from user mode to kernel mode when:					
	A system call occurs	☒	An application opens a text file directly	☐	The user moves the mouse	Memory is automatically garbage-collected
03	Which of the following are advantages of UEFI over BIOS ?					
	Supports disks larger than 2 TB	☒	Limits the system to 4 primary partitions	☐	Supports Secure Boot	☒ Uses GPT instead of MBR ☒
04	Which environment provides the highest isolation between Windows and Linux?					
	WSL	☐	Virtual Machine	☒	Dual-Boot	CUI
05	Which of the following runs Linux directly alongside Windows without a separate OS installation ?					
	Virtual Machine	☐	Dual-Boot	☐	WSL	☒ Bootloader
06	Which setup gives native hardware performance for Linux?					
	WSL	☐	Virtual Machine	☐	Dual-Boot	☒ Container
07	Which of the following features is exclusive to UEFI ?					
	Uses MBR partitioning	☐	Text-only interface	☐	Legacy boot sector reading	Supports Secure Boot ☒
08	Errors occurring in which mode are more likely to crash the operating system?					
	User mode	☐	Guest mode	☐	Kernel mode	☒ Debug mode
09	A TTY device usually represents:					
	a graphical session	☐	a virtual terminal (Ctrl+Alt+F1...F6)	☒	a network socket	a kernel process
10	User mode ensures:					
	Direct access to I/O devices	☐	Protection from faulty applications	☒	Execution of CPU scheduling algorithms	Control of memory management hardware

PART 2: LAB. EXERCISES – LINUX SHELL COMMANDS (07 POINTS)**EXERCISE 1 (2 = 2*1 POINTS) : Linux Command Errors: Identification and Troubleshooting**

For each command below, identify the error it produces, explain why the command fails, and suggest a correct way to fix it.

Commands		Error / Reason for Failure / Corrected Command
01	<code>cp /etc/passwd /root/</code>	Error : cp: cannot create regular file '/root/passwd': Permission denied Explanation : – /root/ is owned by the root user, so a normal user cannot write there. – Requires superuser privileges. Correct ways to fix it : Option 1 — Run the command with root privileges: <code>sudo cp /etc/passwd /root/</code> Option 2 — Switch to root first: <code>sudo -i</code> <code>cp /etc/passwd /root/</code> Option 3 — Copy to a directory where you have permissions: <code>cp /etc/passwd ~/</code>
02	<code>rm test/</code>	Error : rm: cannot remove 'test/': Is a directory Explanation : – rm by default removes files, not directories. – Trying to remove a directory without the recursive flag causes an error. Fix: <code>rm -r test/</code> – r removes the directory and its contents recursively.

EXERCISE 2 (2.5 = 10*0.25 POINTS) : Provide the command needed to produce the expected output for each statement.

Expected Results / Information		Commands
01	Show long format listing (permissions, owner, group, size, modification date, etc.) with inodes	<code>ls -li</code>
02	Identify quickly the nature of each file in Unix/Linux systems	<code>ls -F</code>
03	Quickly find where a command's executable, source, and manual are located on the system.	<code>whereis <command></code>
04	Real-time view of system processes. It shows CPU, memory usage, and running processes, updated continuously.	<code>top</code>
05	Search all files in the /etc directory that contain the string 'root'.	<code>grep root /etc/*</code>
06	Print out the group memberships of a user	<code>groups username</code>
07	Immediately shuts down the system and stops all processes	<code>sudo shutdown -h now</code>
08	Commands that allow you to search the manual page database for entries containing a given keyword	<code>man -k <keyword></code> <code>apropos <keyword></code>
09	Creates or updates an environment variable named VAR_NAME (Visible to current shell and all child processes) and sets it to value.	<code>export VAR_NAME=value</code>
10	Update the list of available packages on the system	<code>sudo apt update</code>

EXERCISE 3 (2.5 = 10*0.25 POINTS) : Understanding Command Behavior: Explanation and Output

Commands	Results / Explanations
01 <code>ls -r -l</code>	Displays a detailed, long-format list of files/directories in reverse order.
02 <code>mkdir -p A/B/C</code>	Create directory C. If A and B does not exist, they are created
03 <code>ls a*</code>	Lists all files and directories in the current directory whose names start with the letter a.
04 <code>ls -lh</code>	Lists files in the current directory in long format with human-readable file sizes (e.g., KB, MB).
05 <code>rm -rf folder_name</code>	Removes the specified directory and all of its contents, including files and subdirectories, without prompting for confirmation.
06 <code>cat /etc/shells</code>	Shows a list of all available shells on the system
07 <code>df -h</code>	Displays disk space usage for all mounted filesystems, with the -h flag providing human-readable output (e.g., in GB or MB).
08 <code>grep "pattern" file.txt</code>	Finds and displays all lines in file.txt that contain the specified pattern
09 <code>echo \$XDG_CURRENT_DESKTOP</code>	Prints the value of the environment variable XDG_CURRENT_DESKTOP. This variable indicates the current desktop environment your session is using, for example: GNOME, KDE, XFCE, LXDE, etc.
10 <code>killall systemd</code>	Close a session : Terminate all running processes named systemd.